

Modelling Volatility Surfaces for Option Pricing

Kanisha Shah

22 July 2026

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Abstract

This report examines the construction and modelling of implied volatility surfaces using observed option market data. After introducing the key concepts of option pricing and implied volatility, a three-dimensional surface is created to describe the relationship between implied volatility, strike price, and time to maturity. The accuracy of this surface is assessed using Leave-One-Out Cross-Validation (LOOCV), to evaluate and minimise prediction error.

A raw Stochastic Volatility Inspired (SVI) parametrisation is then calibrated to the data and tested using the same validation method. The performance of the initial surface and the SVI model is compared to determine which provides a more accurate representation of market-implied volatility. The report highlights practical techniques for fitting volatility surfaces and assesses their effectiveness using real market data.

1 Introduction

Options are financial derivatives whose value depends on the price of an underlying asset. While option-like contracts have existed in various forms for centuries, a proper mathematical theory of option pricing is relatively recent. Early examples are sometimes traced back to stories from ancient Greece, such as accounts involving Thales of Miletus, as well as hedging practices during events like the Dutch tulip mania in the 17th century [5]. However, these were not developed within a formal mathematical framework.

The modern theory really begins with the Black-Scholes model introduced by Black and Scholes (1973). Under a set of modelling assumptions for the underlying asset, the model leads to a partial differential equation for the price of a European option, and in some cases a closed-form solution. The option price depends on several parameters, including the risk-free rate, time to maturity, and in particular, the volatility of the underlying asset.

Volatility is one of the most important of these parameters, since it captures the level of uncertainty in the assets price movements and has a strong effect on option values. Unlike quantities such as the risk-free rate or maturity, volatility cannot be directly observed in the market. Instead, it is typically found from market option prices by inverting the Black-Scholes formula, which leads to the idea of implied volatility.

When implied volatility is calculated across different strike prices and maturities, it is usually not constant. Instead, it shows a skew, giving what is known as the implied volatility surface. This is one of the main flaws of the Black-Scholes model, since the model assumes volatility is constant [9, Chapter 3.10]. To deal with this, more advanced models allow volatility to vary with both the asset price and time, leading to the idea of local volatility.

2 Mathematical Theory

2.1 What are options?

In financial markets, assets such as stocks and options are traded continuously. The price of these assets varies over time in response to new information, such as supply and demand. An option is a *right*, but *not an obligation* to buy an underlying asset by a certain date at a certain price fixed at inception.

The date by which the option must be exercised, T , is called the expiration date or the time to maturity. An option that can only be exercised on the expiration date is called a *European* option. Similarly, an *American* option can be exercised at any time before or on the expiration date.

A *call* option represents the right to buy a particular asset for an agreed amount at a specified time in the future. If we use S_t to represent the underlying stock price and K the strike price, we can form a payoff function to gain a sense of the potential profit to be gained by investing in a certain option:

$$\max(S_T - K, 0)$$

In contrast, a *put* option represents the right to sell a particular asset for an agreed amount at a specified time in the future, with its respective payoff function below:

$$\max(K - S_T, 0)$$

These payoff functions describe the value of an option at maturity. The challenge is determining what that payoff is worth today. This is done by taking the expected payoff and discounting it back to the present [8, Chapter 6.11]:

$$\text{option value} = e^{-r(T-t)} \mathbb{E}[\text{payoff}(S)]$$

Here, r is the risk-free interest rate, and $\text{payoff}(S)$ refers to the payoff functions referred to above, evaluated using the underlying asset price at maturity, S_T .

To calculate this expected payoff, we need to know the probability distribution of the underlying asset price at maturity, S_T . Since this distribution is not known in advance, we introduce a *stochastic model* describing how the asset price evolves over time.

2.2 Geometric Brownian Motion

Stochastic means to involve or follow a random probability distribution or pattern. Stock prices are unpredictable, and the *Geometric Brownian Motion* model can describe this behaviour by combining a predictable pattern with a random component. This is the stochastic differential equation for stock prices

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

where μ measures the expected rate of return, also known as the drift, and σ measures volatility. The random component of the model is represented by the *Wiener process* W_t .

2.3 Wiener Process

The randomness in the *Geometric Brownian Motion* (GBM) model is introduced through the term dW_t , which represents a small increment of a *Wiener process*, also known as *Brownian motion*. A *Wiener process* is a continuous-time stochastic process commonly used to model the unpredictable fluctuations observed in financial markets. Its path evolves continuously over time without making jumps, but it is highly irregular and constantly changing direction.

Over a very small time interval dt , the increment dW_t is normally distributed with mean 0 and variance dt [9, Chapter 2.2], which can also be written as:

$$dW_t \sim N(0, dt)$$

An important property of a *Wiener process* is that it has *independent increments*. If (t_1, t_2) and (t_3, t_4) are non-overlapping time intervals, then the random variables

$$W_{t_2} - W_{t_1} \quad \text{and} \quad W_{t_4} - W_{t_3}$$

are independent. In other words, knowing how the process moved during one time period provides no information about how it will move during a future, non-overlapping period.

This property also makes the Wiener process straightforward to simulate. The accumulation of many small, independent random changes can be written as

$$W_t = \int_0^t dW_s,$$

Equivalently, it may be constructed by summing independent increments over increasingly small time intervals,

$$W_t = \lim_{N \rightarrow \infty} \sum_{i=1}^N (W_{t_i} - W_{t_{i-1}}).$$

This tells us that *Wiener Process* can be regarded as the continuous-time limit of a random walk [2].

2.4 Black-Scholes Differential Equation

Now that we have established the underlying asset dynamics, we can introduce the Black-Scholes model for pricing European options. It assumes that trading is continuous, there are no arbitrage possibilities and the risk-free interest rate

r and volatility σ are known functions of time over the life of an option [9, Chapter 3.5]. Under these assumptions, the value $V(S, t)$ of a European option satisfies the Black-Scholes partial differential equation.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

where $V(S, t)$ represents the value of a European option at time t , where S is the price of the underlying asset.

By solving the Black-Scholes differential equation with the option's payoff at expiry as the boundary condition, the Black-Scholes pricing formula is obtained, which can be used to determine the theoretical value of European options. For a European call option, the price is given by [2]

$$c_t = S_t N(d_1) - K e^{-r(T-t)} N(d_2)$$

where

$$d_1 = \frac{\ln \frac{S_t}{K} + (r + \frac{\sigma^2}{2})(T - t)}{\sigma \sqrt{T - t}}$$

and

$$d_2 = d_1 - \sigma \sqrt{T - t}$$

and $N(x)$ is the cumulative distribution function of the standard normal distribution.

Similarly, the price of a European put option is

$$p_t = K e^{-r(T-t)} N(-d_2) - S_t N(-d_1)$$

The Black-Scholes pricing formula depends on the current asset price, strike price, time to maturity, risk-free interest rate, and the volatility of the underlying asset.

2.5 Implied Volatility

While most of the inputs required by the Black-Scholes model are directly observable in the market, the volatility parameter cannot be observed and must instead be inferred from option prices. When substituted into the Black-Scholes pricing formula, the value of volatility that reproduces an option's observed market price is known as the *implied volatility*.

Implied volatility is widely used as a measure of the market's expectation of future volatility of the underlying asset [8, Chapter 8.9]. In practice, it is obtained by numerically solving the Black-Scholes formula for the volatility parameter σ , using the market price of the option as the target value.

However, implied volatility is not constant across different strike prices and maturities. Instead, market-implied volatilities exhibit systematic patterns that give rise to the *implied volatility surface*. This observation is inconsistent with the Black-Scholes assumption that volatility is constant and identical for all options written on the same underlying asset.

When implied volatility is typically plotted against strike price for a fixed maturity, the resulting curve is often not flat. If implied volatility increases as the strike moves away from the current asset price in either direction, the pattern is known as a *volatility smile* or a *volatility skew*. In the next section, we will create a visual representation of this.

3 Implementation of a Volatility Surface using Geometric Interpolation

Having established why implied volatility varies with strike and maturity, this section builds an actual implied volatility surface from real market data. The surface is constructed directly from observed option prices using geometric interpolation, filling in the gaps between traded strikes and maturities to produce a smooth three-dimensional surface. The accuracy of the interpolated surface is then tested using leave-one-out cross-validation.

3.1 Methodology

The implied volatility surface was created using FTSE 100 index call and put option data from 8 June 2026 to 12 June 2026. The data were collected from Bloomberg Terminal, with each option contract treated as a separate observation. For each contract, the strike price, time to maturity, and implied volatility were used to construct the surface, with strike price on the x-axis, time to maturity on the y-axis, and implied volatility on the z-axis.

Options where the implied volatility was listed as N/A were removed from the dataset. The remaining data were then interpolated to create a smooth three-dimensional implied volatility surface, making it easier to identify and analyse patterns such as the volatility smile and volatility skew. The code was created with the help of Claude (AI) and the process involved several prompts, starting with a general question about how to create a 3D surface plot in Python, before adapting the code to use the specific variables in this study: strike price, time to maturity, and implied volatility. Additional prompts were then used to understand how the code worked, implement the leave-one-out validation approach, and improve the final visualisation of the implied volatility surface.

The data is first separated into call and put options and stored as NumPy arrays. A range is created for each axis, and these are combined using a meshgrid to create a two-dimensional grid. Linear interpolation, using Delaunay

triangulation, is then used to estimate the implied volatility at every point on the grid from the available data.

Since linear interpolation only works within the convex hull of the data [6], nearest-neighbour interpolation is used to fill in the remaining gaps around the edges and corners. Finally, the original data points and the interpolated implied volatility surface are plotted on a three-dimensional graph.

3.2 Results

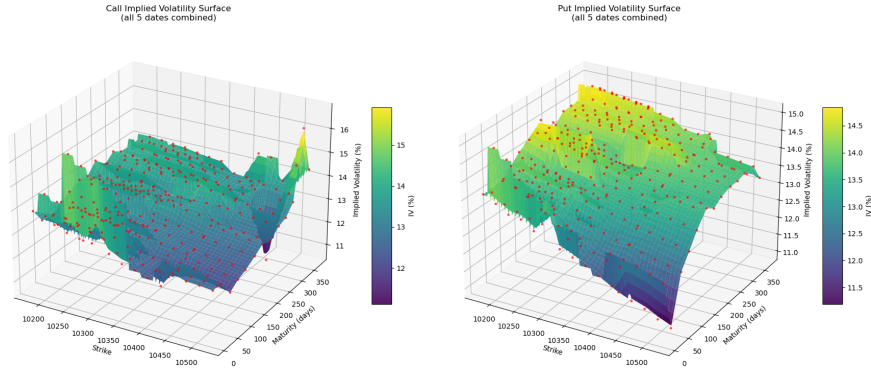


Figure 1: Implied Volatility Surface

Both surfaces display the volatility smile and skew patterns discussed in Section 2.6. The put surface (right-hand side) shows a clear skew, with implied volatility generally higher for shorter maturities and increasing as the strike price decreases. The call surface (left-hand side) is less smooth, with a noticeable dip around a maturity of 250 days and a sharp spike in the region of long maturities and high strikes.

These sharp spikes are unlikely to represent genuine market behaviour. Instead, they are probably caused by limitations in the available data and the interpolation method. Because interpolation accounts for every observed data point exactly, it does not smooth out noisy or unusual quotes. In regions where market data are sparse, particularly near the edges of the strike-maturity grid, isolated observations have a disproportionately large effect on the estimated surface.

The existence of volatility smiles and skews provides strong empirical evidence that the constant-volatility assumption of the Black-Scholes model does not hold in practice. This discrepancy is one of the primary motivations for more sophisticated volatility models.

3.3 Leave-one-out cross-validation

The accuracy of the implied volatility surface was tested by removing one option at a time from the dataset and predicting its implied volatility using the remaining data. The predicted values were then compared with the actual values using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Lower error values suggest that the interpolation method provides a good fit to the observed option data.

This code tests the accuracy of the implied volatility surface by comparing predicted implied volatilities with actual values across all available options. It rebuilds the surface and measures the difference between the estimated and observed implied volatilities, allowing the overall accuracy of the interpolation method to be assessed. The code calculates root mean square error (RMSE) and mean absolute error (MAE) values to summarise the average prediction error for both call and put options. It also identifies the five predictions with the largest errors, showing where the surface has the most difficulty estimating implied volatility.

Call RMSE: 0.5017, MAE: 0.3008

Strike	Maturity	Actual	Predicted	Error	AbsError
10475	308	10.754936	13.588098	2.833162	2.833162
10325	11	12.908485	15.416399	2.507914	2.507914
10350	11	12.628941	15.045585	2.416644	2.416644
10525	343	16.342356	13.999671	-2.342685	2.342685
10250	8	16.615707	14.424156	-2.191551	2.191551

Put RMSE: 0.3662, MAE: 0.2092

Strike	Maturity	Actual	Predicted	Error	AbsError
10425	375	13.990518	12.282960	-1.707558	1.707558
10200	9	14.960991	13.423443	-1.537548	1.537548
10200	10	13.423443	14.925418	1.501975	1.501975
10225	10	13.209211	14.694120	1.484909	1.484909
10525	35	11.030149	12.419516	1.389367	1.389367

The results show that the implied volatility surface was able to predict missing values with a good level of accuracy. Across 375 call options, the RMSE was 0.5017 percentage points, while put options had a slightly lower RSME of 0.3662 percentage points. The largest errors were found for options with unusual strike prices or maturities, where there were fewer similar options available to estimate from. Despite these larger errors, most predictions were close to the actual implied volatilities, showing that the surface provides a reasonable representation of the observed option data.

While interpolation was used to create a smooth surface and fill in the gaps between observed data points, this approach does not necessarily enforce arbitrage-free conditions. In the next section, we explore the implementation of stochastic volatility models to provide a more financially consistent framework.

4 SVI Parametrisation

As we have seen before, for option prices with the same expiry but different strikes, implied volatility is not constant. Instead, it typically forms a smile or skew shape. *Stochastic Volatility Inspired* (SVI) is simply a five-parameter curve designed to fit that shape well, for one expiry at a time. The raw SVI form is given by [4, Section 3.1] as

$$w(k; \chi_R) = a + b \left\{ \rho(k - m) + \sqrt{(k - m)^2 + \sigma^2} \right\}, \quad \chi_R = \{a, b, \rho, m, \sigma\},$$

where $k = \ln(K/F)$ is log-moneyness relative to the forward price F , and $w(k, T) = \sigma_{BS}(k, T)^2 T$ is the total implied variance.

Each of the five parameters has its own role:

- a shifts the whole curve up or down (overall variance level)
- b controls how steeply the curve rises on both sides (wing steepness)
- ρ tilts the curve left or right (skew/asymmetry)
- m shifts the curve's minimum sideways
- σ controls how rounded or sharp the curve is near its minimum (curvature)

What is moneyness?

Moneyness describes how a strike price compares with the current market level of the underlying asset. An option is *at-the-money* (ATM) when its strike price is very close to the current price (or forward price) of the asset. If exercising the option right now would already make money, it's *in-the-money* (ITM); if it would lose money, it's *out-of-the-money* (OTM) [7].

4.1 Methodology

For each expiry, market strikes K_i and mid implied volatilities were converted into total variance observations $w_i = \sigma_{BS,i}^2 T_i$.

Before we can compute log-moneyness, we need the forward price F_T , since k is measured relative to it, not to the raw strike. This was recovered using *put-call parity*: for a call and put on the same strike and expiry, the difference between their prices depends only on the forward and a discount factor D ,

$$C_{mid}(K) - P_{mid}(K) = D(F_T - K).$$

This is a straight line in K , so plotting the call-put price difference against strike and fitting a line gives D as the slope and F_T from the intercept. Once F_T is known, log-moneyness is $k_i = \ln(K_i/F_T)$ for every strike.

Now that we know k_i and w_i , the five SVI parameters (a, b, ρ, m, σ) can be fitted separately for each expiry by minimising the sum of squared errors between the market total variance and the model curve,

$$\text{SSE}(a, b, \rho, m, \sigma) = \sum_i (w_i - \hat{w}(k_i; a, b, \rho, m, \sigma))^2.$$

Four constraints were imposed so the fitted curve stays economically sensible: $b \geq 0$ and $\sigma > 0$ (so the curve doesn't fold back on itself), $|\rho| < 1$ (so it doesn't tilt past vertical), and $a + b\sigma\sqrt{1-\rho^2} \geq 0$, which guarantees the variance $w(k)$ never goes negative anywhere on the curve [4, Section 3.1].

4.2 Results

Claude (AI) was used to help write the code for this section. It was prompted to parse the option chain data, estimate the forward price for each maturity using put-call parity, and calibrate the raw SVI parameters (a, b, ρ, m, σ) for each expiry by minimising squared errors while keeping the fit within the model's no-arbitrage constraints. It was also prompted to set up leave-one-out cross-validation to check the accuracy of the model.

Table 1: Raw SVI slice calibration results, UKX option chain (8-June-2026). Parameters (a, b, ρ, m, σ) fit by minimizing $\sum_i (w_i - \hat{w}(k_i; a, b, \rho, m, \sigma))^2$ per maturity, with $\hat{w}(k) = a + b\{\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2}\}$. $\text{RMSE}(w)$ is measured on the total variance scale; $\text{RMSE}(\text{vol})$ is the same fit error converted to the implied volatility scale via $\sigma = \sqrt{w/T}$, which is comparable across maturities.

Days	T (yrs)	F	a	b	ρ	m	σ	$\text{RMSE}(w)$	$\text{RMSE}(\text{vol})$
11	0.030	10383.1	-0.00936	0.09903	0.0000	0.0036	0.0991	5.93e-06	8.06e-04
39	0.107	10408.8	-0.00817	0.09909	0.0000	0.0012	0.0991	2.51e-05	9.41e-04
74	0.203	10388.6	-0.00677	0.09984	-0.0013	0.0170	0.0986	9.96e-07	1.94e-05
102	0.279	10415.1	-0.00539	0.10009	-0.0018	0.0188	0.0985	6.83e-06	9.39e-05
130	0.356	10435.0	-0.00478	0.10117	-0.0041	0.0401	0.0974	1.41e-05	1.50e-04
165	0.452	10453.0	-0.00359	0.10196	-0.0054	0.0512	0.0967	1.25e-05	1.05e-04
193	0.529	10478.8	-0.00098	0.10123	-0.0036	0.0312	0.0977	6.65e-06	4.62e-05
221	0.605	10515.7	0.00143	0.09957	-0.0000	-0.0182	0.0996	1.45e-05	8.76e-05
256	0.701	10537.2	0.00412	0.10097	-0.0025	0.0141	0.0985	3.61e-06	1.78e-05
284	0.778	10519.3	0.00328	0.10217	-0.0050	0.0412	0.0972	2.16e-05	1.02e-04
312	0.855	10524.8	0.00268	0.10365	-0.0082	0.0723	0.0955	3.92e-06	1.68e-05
347	0.951	10527.4	0.00621	0.10261	-0.0058	0.0481	0.0970	4.43e-06	1.68e-05
375	1.027	10542.7	0.00713	0.10304	-0.0066	0.0541	0.0966	7.18e-06	2.53e-05
403	1.104	—	—	—	—	—	—	—	failed ^a
438	1.200	—	—	—	—	—	—	—	failed ^a

^a Forward F could not be estimated: too few strikes had both a call and a put mid-price quote to run the put-call-parity regression.

Table 1 shows the calibrated SVI parameters for the FTSE 100 (UKX) option chain as of 8 June 2026, across 15 maturities (expiry dates) from 11 to 438 days. Thirteen of the fifteen slices were fit successfully. The $\text{RMSE}(w)$ column measures the error in total variance, and $\text{RMSE}(\text{vol})$, which converts that same error into implied volatility so it can be fairly compared across maturities.

Looking at $\text{RMSE}(\text{vol})$ across the term structure, we can see that the error is largest for the shortest-dated options and steadily shrinks as maturity increases. At 11 days, $\text{RMSE}(\text{vol})$ is around 8×10^{-4} , and it stays relatively high through the 39-day maturity as well, before dropping sharply for the 74-day expiry and beyond. From around 100 days onward, $\text{RMSE}(\text{vol})$ settles into a much smaller range, mostly between 1×10^{-5} and 1.5×10^{-4} , with the longest maturities (312, 347, and 375 days) among the best fits in the whole table.

This pattern makes sense given how implied volatility smiles typically behave. Short-dated smiles tend to be steeper and more sensitive to small changes in strike, so there is more curvature for the model to match exactly, and any small mismatch in shape shows up as a larger volatility error once we convert back from total variance. Longer-dated smiles are generally flatter and smoother, which is an easier shape for SVI’s functional form to reproduce almost exactly, which is why we can see smaller errors from 100 days onward.

4.3 Leave-one-out cross-validation

The RMSE values in Table 1 only measure how well the model matches the data used for calibration. Since each maturity contains five data points and the SVI model has five parameters, the model can fit those points almost exactly. To assess predictive performance, we applied *leave-one-out cross-validation* (LOOCV). Each strike was removed in turn, the model was re-fitted to the remaining four strikes, and then used to predict the omitted point. The LOOCV RMSE is the average error across the five held-out predictions.

The in-sample RMSE column, is simply the calibration error from Table 1, now expressed on the same implied volatility scale, so it can be compared directly to the LOOCV column next to it.

Table 2: Leave-one-out cross-validation of the per-maturity raw SVI fits, UKX option chain (8-June-2026). For each maturity, each of the n strikes is held out in turn, the remaining $n - 1$ points are used to refit (a, b, ρ, m, σ) , and the held-out point is predicted. Both in-sample and LOOCV RMSE are measured on the implied volatility scale ($\sigma = \sqrt{w/T}$) rather than total variance, so results are directly comparable across maturities.

Days	T (yrs)	n	In-sample RMSE(vol)	LOOCV RMSE(vol)	Ratio
11	0.030	5	8.06e-04	1.72e-03	2.14
39	0.107	5	9.41e-04	1.34e-03	1.42
74	0.203	5	1.94e-05	3.20e-05	1.65
102	0.279	5	9.39e-05	1.84e-04	1.96
130	0.356	5	1.50e-04	2.72e-04	1.80
165	0.452	5	1.05e-04	2.17e-04	2.07
193	0.529	5	4.62e-05	7.83e-05	1.69
221	0.605	5	8.76e-05	1.42e-04	1.62
256	0.701	5	1.78e-05	3.18e-05	1.78
284	0.778	5	1.02e-04	1.83e-04	1.80
312	0.855	5	1.68e-05	3.61e-05	2.16
347	0.951	5	1.68e-05	2.95e-05	1.75
375	1.027	5	2.53e-05	4.20e-05	1.66
403	1.104	5	—	—	<i>skipped</i> ^a
438	1.200	5	—	—	<i>skipped</i> ^a

^a Forward could not be estimated for this maturity (see main calibration table), so no LOOCV was run.

Across every maturity, the LOOCV error is slightly larger than the in-sample error, typically by a factor of around 1.5 to 2. This makes sense as each maturity only has five strikes, and the SVI model has five parameters to fit. That means, in-sample, the model has just barely enough flexibility to pass through all five points almost exactly. When we remove one point, the error only grows moderately, rather than blowing up, suggesting the fitted smile shape is reasonably stable and that the model generalises relatively well.

5 Local Volatility

As we have seen before, volatility σ is not single constant, the same for every strike and every maturity. In reality this does not hold: options on the same underlying, but with different strikes and maturities, trade at different implied volatilities. One way to keep using a Black-Scholes-style pricing framework while still matching these observed prices is to let volatility depend on the underlying's level and on time, $\sigma = \sigma(S, t)$, instead of treating it as one fixed number [3, Chapter 1]. This is the idea behind *local volatility*.

5.1 Dupire’s Local Volatility Model

Dupire showed that it is possible to construct a local volatility function, $\sigma_{loc}(S, t)$, that exactly matches all observed European option prices across different strikes and maturities. Under this model as shown in [3], the underlying asset follows

$$dS_t = rS_t dt + \sigma(S_t, t) S_t dW_t,$$

which is very similar to the Black-Scholes model. The key difference is that volatility is no longer fixed and changes depending on the current asset price and the time.

The local volatility function can be recovered directly from the market’s call price surface $C(K, T)$ using *Dupire’s equation* [1]:

$$\sigma_{loc}^2(K, T) = \frac{\frac{\partial C}{\partial T} + rK \frac{\partial C}{\partial K}}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}}.$$

The benefit of local volatility is that it incorporates the smile and skew seen in market implied volatilities. This lets us build a model that is consistent with observed vanilla option prices and can also be used to price more complex derivatives.

Overall, local volatility provides a way to turn the information contained in the implied volatility surface into a full model for the underlying asset dynamics. While SVI is useful for fitting a smooth and realistic volatility surface to market data, the local volatility model uses that fitted surface to describe how the asset price evolves over time.

6 Conclusion

This report investigated the modelling of implied volatility surfaces using observed option market data. After introducing the key concepts of option pricing and implied volatility, a three-dimensional volatility surface was constructed as a function of strike price and time to maturity. The accuracy of this surface was assessed using Leave-One-Out Cross-Validation (LOOCV), providing a measure of its ability to represent unseen observations.

A raw SVI parametrisation was then calibrated to the same data and evaluated using the same validation framework. The results showed that the SVI model significantly outperformed the interpolation-based approach, producing substantially lower prediction errors while maintaining a smooth and realistic representation of the volatility surface.

Finally, local volatility and Dupire’s framework were briefly discussed as an

extension of volatility surface modelling. Overall, the report demonstrates that while interpolation can provide a useful initial representation of market data, parametric approaches such as SVI offer a more accurate and robust framework for modelling implied volatility surfaces. This highlights the importance of combining theoretical models with empirical validation when analysing option markets.

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